Why 96% of Customers Never Return

A Customer Retention Strategy for Olist Fashion

Executive Summary

Olist's overall marketplace performance showed strong year-over-year growth in 2018. However, growth momentum began to flatten despite continued customer acquisition.

A deeper retention analysis revealed that only 3.61% of customers made a second purchase within 180 days, indicating a severe customer retention challenge.

This project investigates:

1. Who are Olist Fashion customers?
2. Why do they fail to return?
3. Which operational factors most strongly influence repurchase behavior?
4. What retention strategies should Olist prioritize?

Using SQL, Python, Machine Learning, NLP, and Tableau, I identified customer segments, quantified churn drivers, and proposed segment-specific retention strategies.

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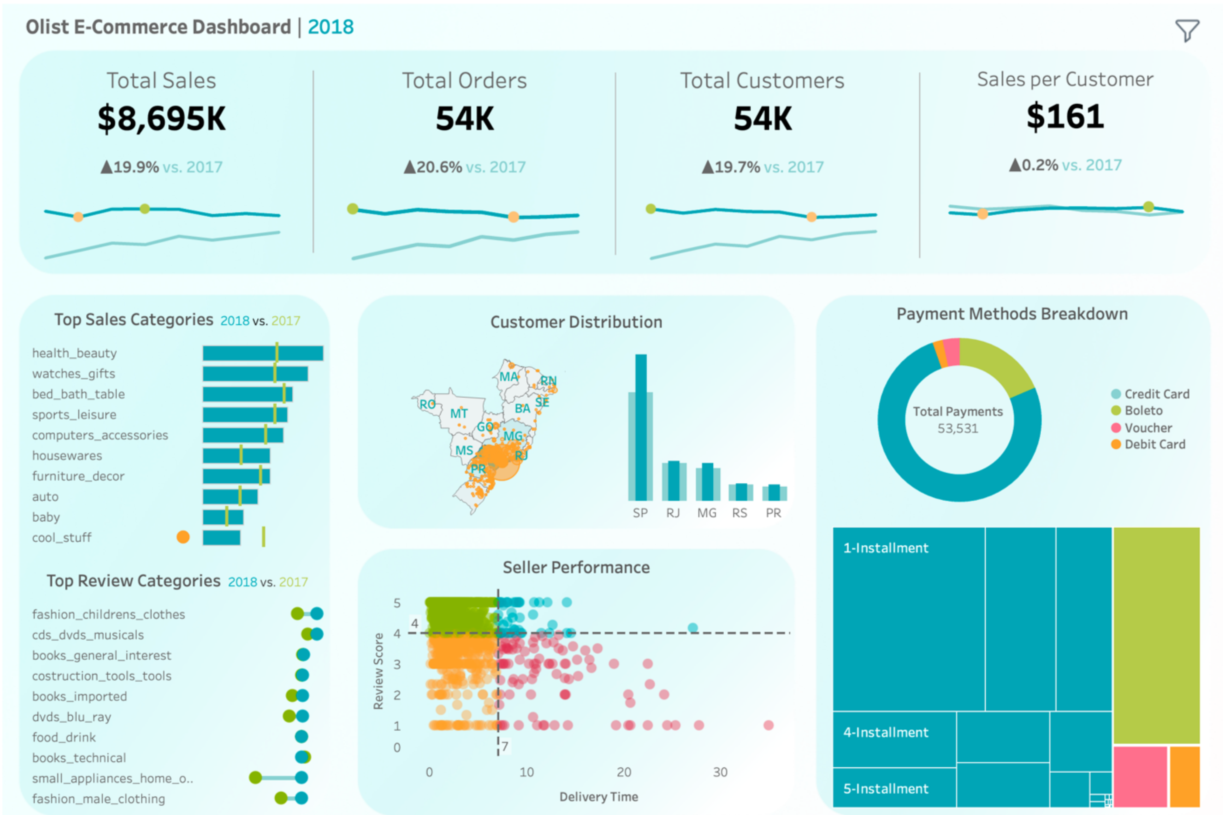
1. Business Context

Marketplace Growth Overview

Using Tableau, I built an executive dashboard covering:

- Revenue
- Orders
- Customer Growth
- Geographic Distribution
- Category Performance
- Seller Performance
- Payment Behavior



The interactive dashboard can be found here.

Key Finding

Although Olist achieved strong YoY growth from 2017 to 2018, growth rates across sales, customers, and orders began to stabilize.

This raised an important business question:

> Is Olist relying too heavily on acquiring new customers while failing to retain existing ones?

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2. Diagnosing the Retention Problem

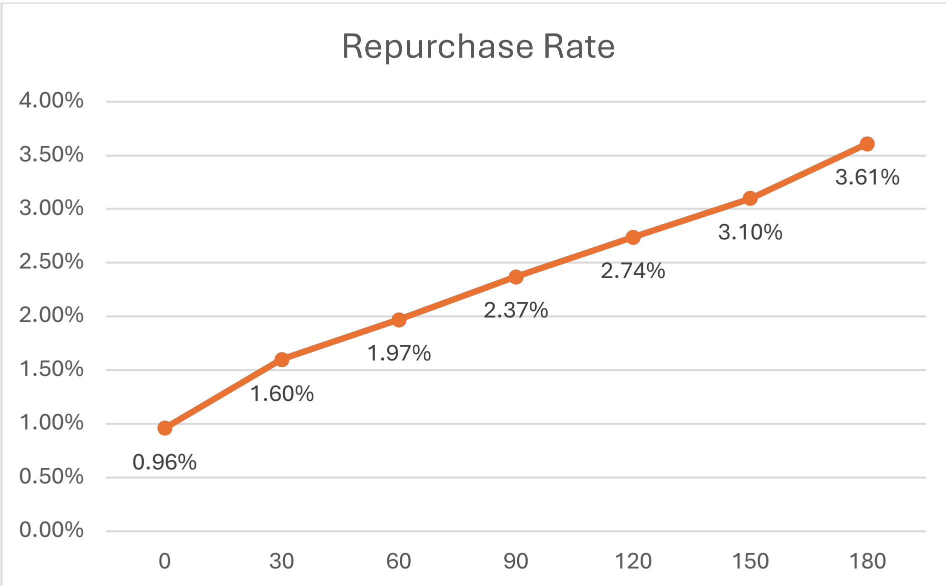
Measuring Repurchase Rate

To evaluate customer retention, I calculated cumulative repurchase rates after a customer's first purchase.

Results

Repurchase within:

- 30 Days: 1.60%
- 90 Days: 2.37%
- 180 Days: 3.61%



Key Finding

96.39% of customers never returned within six months.

This suggests a classic "Leaky Bucket" problem:

Olist continues acquiring customers, but very few remain active long enough to generate repeat revenue.

Business Implication

Customer retention represents a larger growth opportunity than customer acquisition.

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3. Understanding Who the Customers Are

Customer Segmentation Using RFM + K-Means

To understand customer behavior, I created:

- Recency
- Frequency

- Monetary

features for every Fashion customer.

After scaling and preprocessing the data, I applied K-Means clustering.

Resulting Segments

Cluster 0 — One-Timers

Single purchase customers with minimal engagement.

Cluster 1 — Newcomers

Recently acquired customers with limited purchasing history.

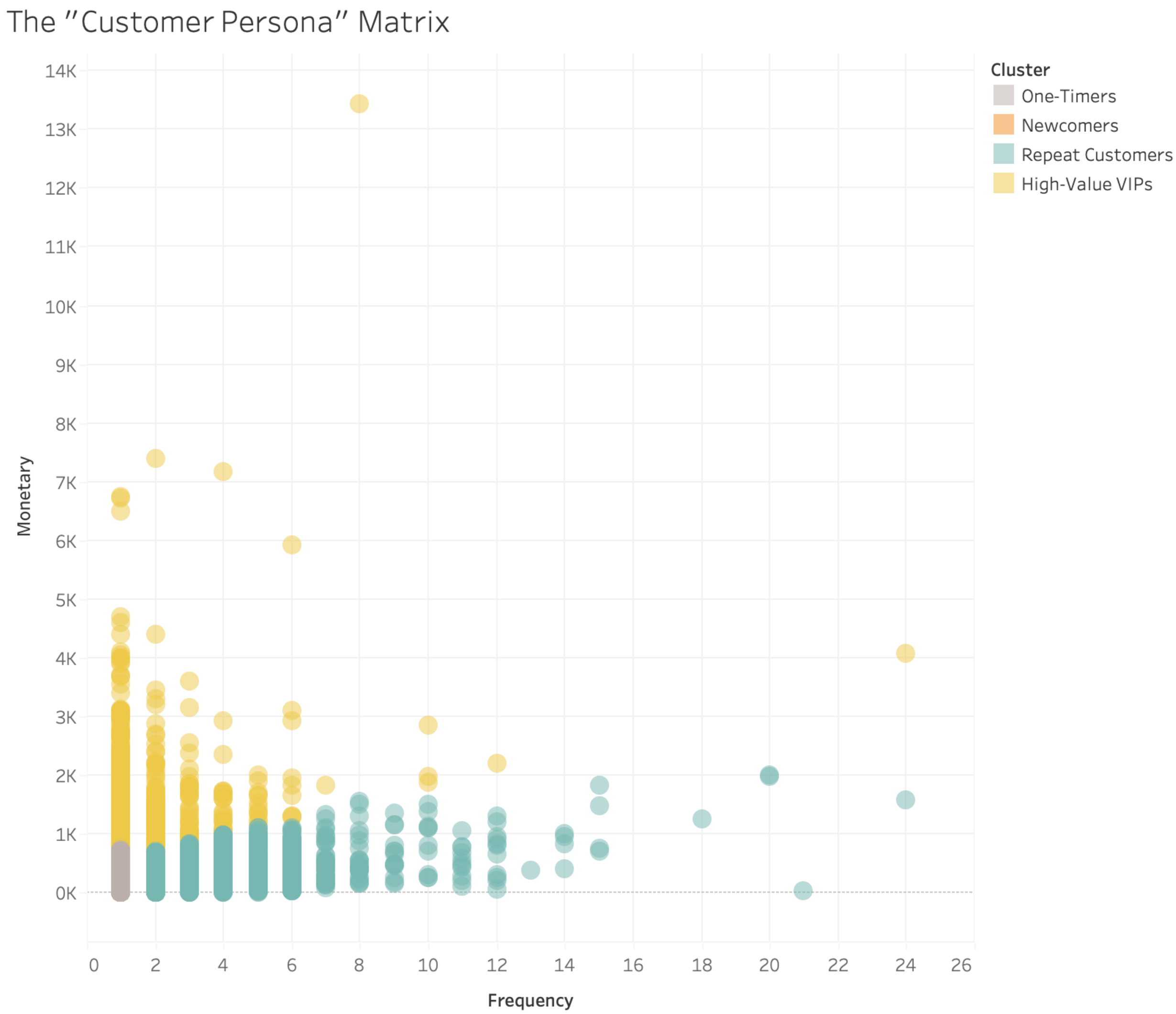
Cluster 2 — Repeat Customers

Frequent buyers with relatively low spending per order.

Cluster 3 — High-Value VIPs

Customers with the highest spending and strongest revenue contribution.

	cluster	User_Count	recency	frequency	monetary
0	0	34719	394.004666	1.049627	110.793336
1	1	46938	129.798628	1.000000	108.462649
2	2	9655	193.260383	2.574625	199.542394
3	3	2046	235.847996	1.271750	1180.645709



Key Finding

Across all four segments, average recency remained high.

This suggests that customer churn is not isolated to one segment.

Retention issues exist throughout the customer base.

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4. Investigating Why Customers Leave

Initial Hypothesis

Two operational factors were suspected:

1. Delivery delays

2. Shipping costs

To identify which variables most strongly influence repurchase behavior, I trained a Decision Tree model.

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5. Discovery of Two Distinct Customer Economies

Decision Tree Findings

The model identified an important behavioral split at:

Average Order Value = $98.71

This threshold separated customers into two economically different groups:

Value Segment

Order Value < $98.71

Premium Segment

Order Value ≥ $98.71

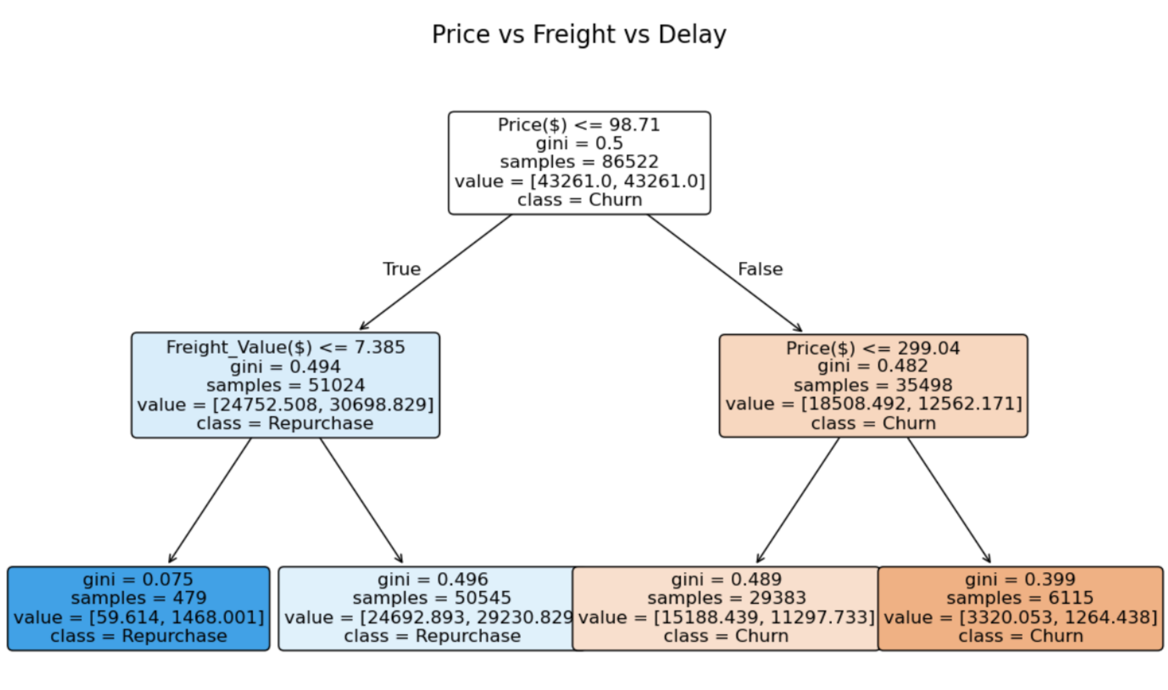
This finding became the foundation for all subsequent churn analysis.

Additional Finding

For low-order-value customers:

Freight Cost < $7.38 significantly increased repurchase probability.

This suggests that shipping costs may be acting as a psychological barrier to repeat purchases.



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6. Quantifying Churn Drivers

Segmented Logistic Regression

Instead of treating all customers as one population, separate Logistic Regression models were built for:

- Value Segment
- Premium Segment

This allowed the impact of logistics and pricing variables to be measured independently.

Value Segment Analysis

Variables:

- Product Price
- Freight Cost
- Delivery Delay

```
                           Logit Regression Results
==============================================================================
Dep. Variable:         is_repurchased   No. Observations:                51024
Model:                          Logit   Df Residuals:                    51020
Method:                           MLE   Df Model:                            3
Date:                Sat, 30 May 2026   Pseudo R-squ.:                0.001975
Time:                        13:28:01   Log-Likelihood:                -21849.
converged:                       True   LL-Null:                       -21892.
Covariance Type:            nonrobust   LLR p-value:                 1.249e-18
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -1.4351      0.035    -40.646      0.000      -1.504      -1.366
price            -0.0042      0.001     -7.822      0.000      -0.005      -0.003
freight_value    -0.0037      0.002     -2.142      0.032      -0.007      -0.000
is_delayed       -0.1469      0.053     -2.758      0.006      -0.251      -0.043
=================================================================================
```

Findings

Lower product prices increase repurchase probability.

Lower freight costs increase repurchase probability.

Delivery delays reduce repurchase probability.

Interpretation

These customers are highly price sensitive.

Retention strategies should focus on affordability and shipping incentives.

Premium Segment Analysis

Variables:

- Product Price
- Freight Cost
- Delivery Delay

```
                           Logit Regression Results
==============================================================================
Dep. Variable:         is_repurchased   No. Observations:                35498
Model:                          Logit   Df Residuals:                    35494
Method:                           MLE   Df Model:                            3
Date:                Sat, 30 May 2026   Pseudo R-squ.:                0.007020
Time:                        13:28:02   Log-Likelihood:                -10697.
converged:                       True   LL-Null:                       -10772.
Covariance Type:            nonrobust   LLR p-value:                 1.417e-32
=================================================================================
                    coef    std err          z      P>|z|      [0.025      0.975]
---------------------------------------------------------------------------------
const            -2.0916      0.035    -60.162      0.000      -2.160      -2.023
price            -0.0012      0.000     -9.483      0.000      -0.001      -0.001
freight_value     0.0025      0.001      2.670      0.008       0.001       0.004
is_delayed       -0.4026      0.081     -4.942      0.000      -0.562      -0.243
=================================================================================
```

Findings

Product price has limited influence.

Freight cost has limited influence.

Delivery delay becomes the strongest predictor of churn.

Sensitivity to delivery delays is approximately three times greater than in the Value Segment.

Interpretation

Premium customers purchase experiences, not discounts.

Operational excellence matters more than pricing incentives.

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7. Testing an Alternative Explanation

Are High-Value Customers Simply Stockpiling?

One alternative explanation was that high-value customers buy in bulk and therefore naturally take longer to repurchase.

To test this hypothesis:

I measured the relationship between basket size and repurchase behavior.

```
               basket_size  is_repurchased
basket_size       1.000000        0.626765
is_repurchased    0.626765        1.000000
```

Finding

Basket size showed a positive relationship with repurchase.

Customers who purchased more items were actually more likely to return.

Conclusion

High-value customers are not stockpiling inventory.

Their churn must be explained by other factors, most notably service experience and delivery performance.

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8. Segment Validation

To validate previous findings, I compared operational metrics across customer clusters.

Repeat Customers

- Lowest average order value
- Lowest freight cost
- Fastest delivery performance
- Highest purchase frequency

High-Value VIPs

- Highest spending
- Greater tolerance for shipping costs
- Strong dependence on delivery reliability

Conclusion

Observed customer behavior aligns closely with findings from the Logistic Regression analysis.

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9. Voice of Customer Analysis

NLP Review Mining

To understand customer sentiment, I analyzed review text from:

- Repeat Customers
- High-Value VIPs

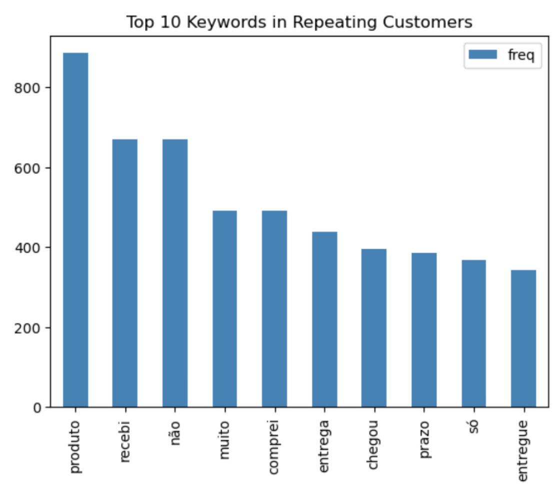
Using keyword extraction and text preprocessing.

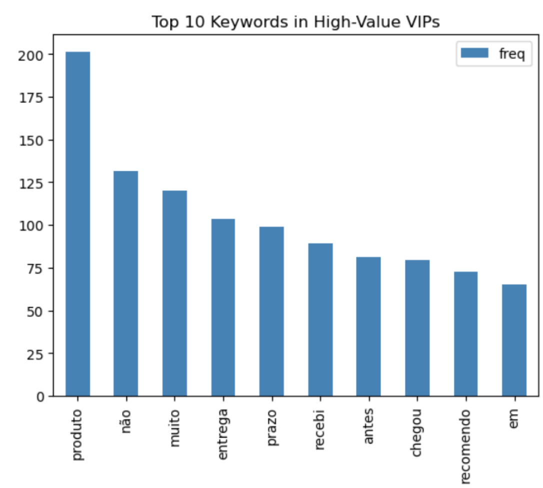
Findings

Delivery-related complaints appeared frequently across both segments.

Common themes:

- Delivery delays
- Shipping experience
- Order arrival timing





Portuguese Keyword	English Meaning
produto	Product
recebi	Received
não	No / Not
muito	Very
comprei	Bought
entrega	Delivery

prazo	Deadline / Term
chegou	Arrived
entregue	Delivered
antes	Before
recomendo	Recommend

Additional Insight

High-Value VIPs frequently used recommendation-oriented language.

Examples:

- Recommend

Business Interpretation

VIP customers represent both:

- Revenue value
- Referral value

Retaining them produces a disproportionately large business impact.

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10. Strategic Recommendations

Strategy 1 — Value Segment

Objective:

Increase repurchase through cost reduction.

Recommended Actions:

- Free Shipping A/B Test
- Shipping Subsidies
- Bundled Promotions
- First Repeat Purchase Coupons

Success Metric:

90-Day Repurchase Rate

Strategy 2 — Premium Segment

Objective:

Increase retention through service quality.

Recommended Actions:

- Priority Fulfillment
- VIP Shipping Program
- Delivery ETA Accuracy
- Premium Membership Benefits

Success Metric:

VIP Retention Rate

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11. Operational Early-Warning Framework

Based on the visual distribution of fulfillment latency across customer segments, I identified potential early-warning zones for retention risk.

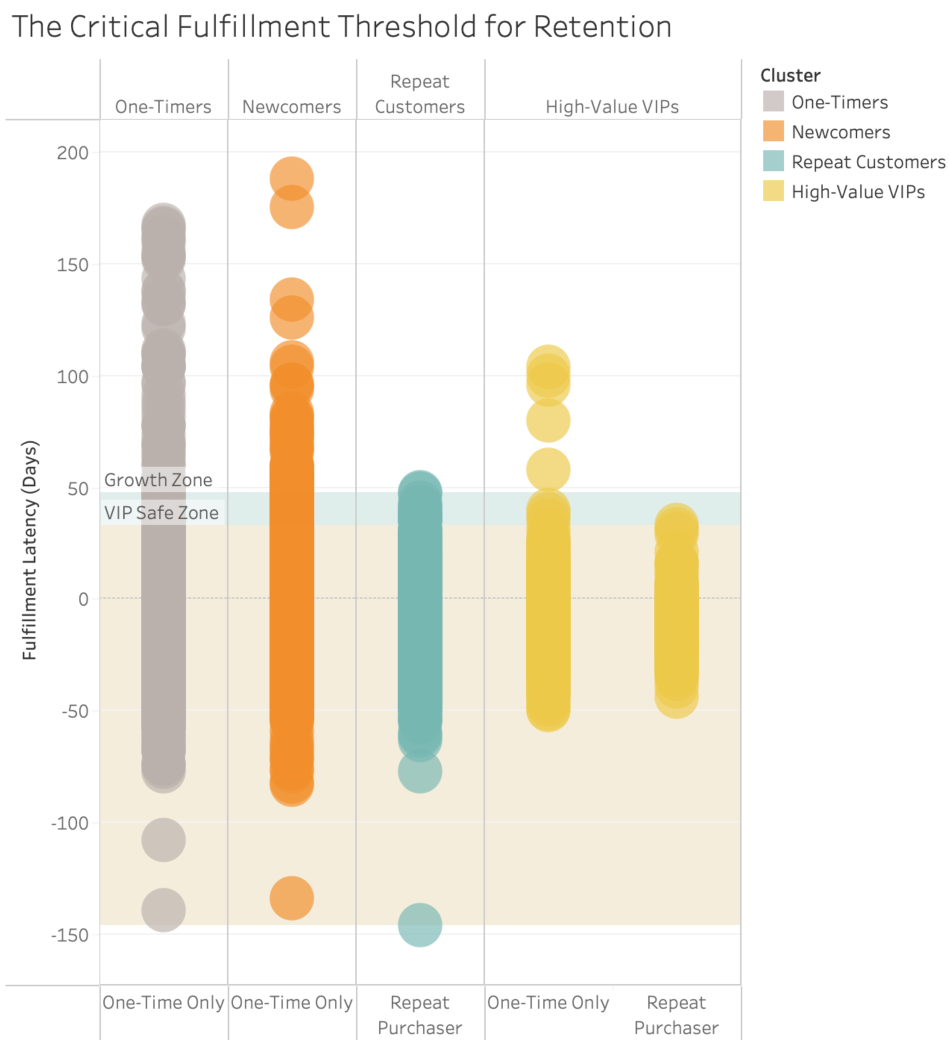
General Customer Base

Fulfillment latency appears to become increasingly risky when delays move beyond approximately 48 days.

High-Value VIP Customers

The risk zone appears earlier, around approximately 33 days, suggesting that premium customers may have a lower tolerance for delivery uncertainty.

These thresholds should not be interpreted as strict causal cutoffs. Instead, they can serve as exploratory operational benchmarks for monitoring fulfillment performance and prioritizing customer recovery actions.



Business Implication

Olist should use fulfillment latency as a retention-risk signal.

When an order approaches the risk zone, the platform could trigger proactive service recovery actions, such as:

- Delivery status updates
- Apology messages
- Customer service escalation
- Shipping fee credits
- VIP priority support

This framework turns delivery delay from a passive operational metric into an active customer-retention trigger.

Final Takeaway

Olist Fashion does not suffer from a customer acquisition problem.

It suffers from a customer retention problem.

For low-value customers, retention is driven primarily by price and shipping costs.

For high-value customers, retention is driven primarily by service quality and delivery reliability.

A one-size-fits-all retention strategy would be ineffective.

The most effective path forward is segment-specific retention management.

Project Assets

Technical repository includes:

1. SQL Scripts

- Data cleaning

- Repurchase rate calculation

- RFM feature engineering

2. Python Notebooks

- Data preprocessing

- K-Means clustering

- Decision Tree threshold analysis

- Segmented Logistic Regression

- NLP review mining

3. Tableau Workbook

- Interactive marketplace performance dashboard

- Customer segmentation views

- Retention and fulfillment-risk analysis views

4. Methodology Notes

- Feature definitions

- Modeling assumptions

- Business interpretation

GitHub Link: https://github.com/chloeliu-analytics/olist-fashion-retention-analysis